

Stochastic Auto Reserving

Minseok Han

2026-10-08

```
conn <- dbConnect(RSQLite::SQLite(), "CommAuto.db")

query <- "
  SELECT AccidentYear, DevelopmentLag, SUM(CumPaidLoss) as CumPaidLoss
  FROM commercial_auto
  WHERE AccidentYear + DevelopmentLag <= 2008 -- Mask the future
  GROUP BY AccidentYear, DevelopmentLag
"
industry_data <- dbGetQuery(conn, query)

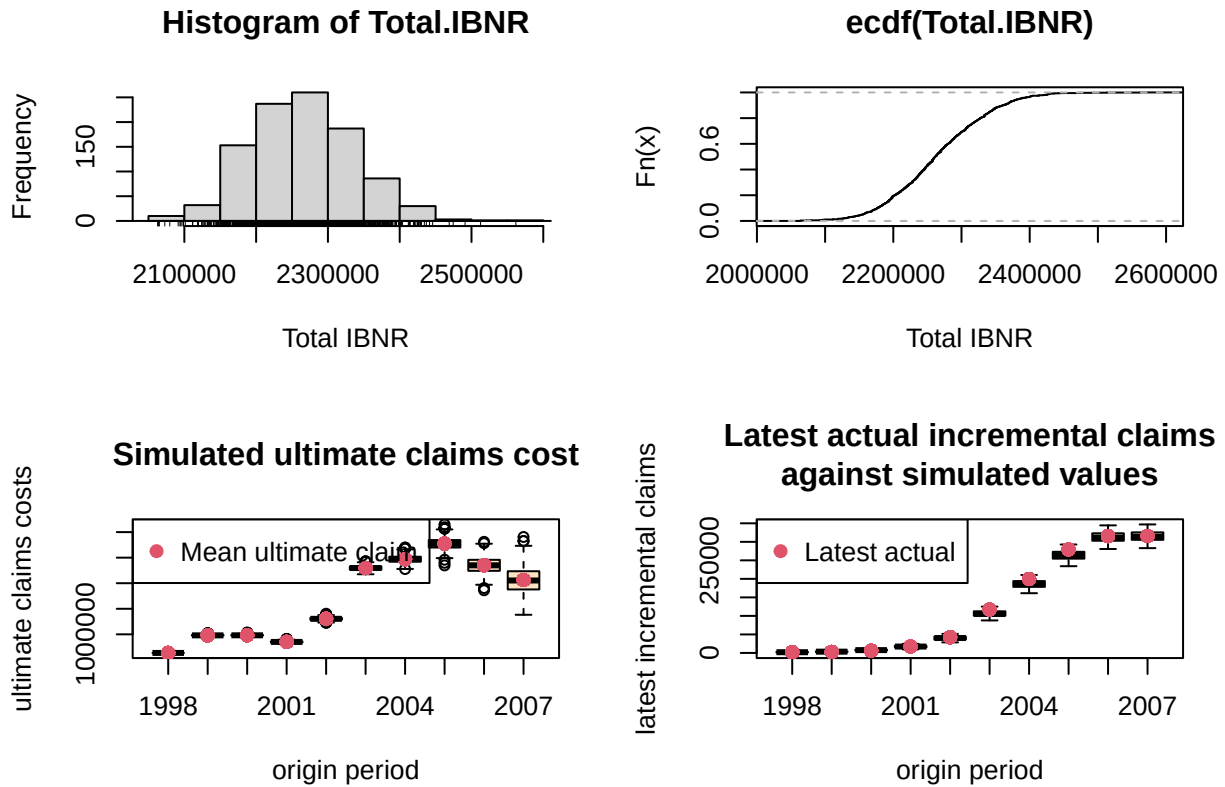
dbDisconnect(conn)

tri_matrix <- dcast(industry_data, AccidentYear ~ DevelopmentLag, value.var = "CumPaidLoss")
tri <- as.triangle(as.matrix(tri_matrix[, -1]))
dimnames(tri)[[1]] <- tri_matrix$AccidentYear

set.seed(88125)
boot_model <- BootChainLadder(tri, R = 1000, process.distr = "gamma")
bootstrap_summary <- summary(boot_model)
print(bootstrap_summary)
```

```
## $ByOrigin
##      Latest Mean Ultimate   Mean IBNR    SD IBNR    IBNR 75%    IBNR 95%
## 1998  927164      927164.0      0.000      0.000      0.000      0.000
## 1999  994075      996232.4    2157.369    1719.315    3147.673    5323.723
## 2000  990905      996482.0    5577.043    2573.815    7107.844   10156.958
## 2001  957567      970407.0   12840.006    3433.866   14902.712   19008.654
## 2002 1027941     1060859.8   32918.824    5458.663   36331.681   42442.169
## 2003 1172557     1259365.7   86808.697    8734.893   92836.378  101493.319
## 2004 1095213     1294038.7  198825.696   13923.690  208486.156  222026.649
## 2005  951424     1354526.0  403102.003   21756.056  417477.383  439711.282
## 2006  644483     1270711.8  626228.819   30151.517  647113.065  678876.785
## 2007  315761     1212718.8  896957.768   49891.472  931135.564  981085.435
##
## $Totals
##              Totals
## Latest:      9077090.00
## Mean Ultimate: 11342506.22
## Mean IBNR:    2265416.22
## SD IBNR:      71864.55
## Total IBNR 75%: 2314173.62
## Total IBNR 95%: 2384477.40
```

```
plot(boot_model)
```



```
# summary <- as.data.frame(bootstrap_summary$Totals)

# wb <- loadWorkbook("Commercial_Auto_Reserving_Dashboard.xlsx")
# if (!"Bootstrap Summary" %in% names(wb)) addWorksheet(wb, sheetName = "Bootstrap Summary")
# writeData(wb, sheet = "Bootstrap Summary", x = summary, rowNames = TRUE)
# saveWorkbook(wb, "Commercial_Auto_Reserving_Dashboard.xlsx", overwrite = TRUE)
```